

Week 4 Day 3 Summary

Feature Engineering, Cross-Validation & Model Comparison

1. Engineered Features

Eight new features were created using only information available within each individual record, ensuring that no information leakage occurred.

Feature	Type	Creation Rule	Justification
age_bucket	Categorical	Group age into ranges	Captures non-linear effects of age on income.
hours_bucket	Categorical	Bin weekly working hours	Differentiates part-time, full-time and overtime workers.
has_capital_gain	Binary	$\text{capital_gain} > 0$	Indicates whether a person has any investment income.
has_capital_loss	Binary	$\text{capital_loss} > 0$	Indicates whether a person reported capital loss.
log_capital_gain	Numeric	$\log(\text{capital_gain} + 1)$	Reduces the skewness of the highly right-skewed capital gain distribution.
log_capital_loss	Numeric	$\log(\text{capital_loss} + 1)$	Reduces the skewness of the capital loss distribution.
higher_education	Binary	$\text{education_num} \geq 13$	Identifies individuals with Bachelor's degree or higher.
education_hours_interaction	Numeric	$\text{education_num} \times \text{hours_per_week}$	Captures the combined effect of education and working hours.

These engineered features were integrated into the preprocessing pipeline using a **FunctionTransformer**, ensuring they were created automatically within every cross-validation fold.

2. Cross-Validated Model Comparison

Five-fold **Stratified Cross-Validation** was used to compare five classification models using identical preprocessing.

Model	Precision	F1 Score	ROC AUC	CV Time (s)
Logistic Regression	0.7461 ± 0.0062	0.6765 ± 0.0101	0.9131 ± 0.0019	2.32
Random Forest	0.7140 ± 0.0074	0.6652 ± 0.0059	0.8956 ± 0.0018	23.74
HistGradientBoosting	0.7783 ± 0.0085	0.7087 ± 0.0083	0.9281 ± 0.0018	12.33
XGBoost	0.7628 ± 0.0079	0.7053 ± 0.0075	0.9253 ± 0.0029	5.43
LightGBM	0.7755 ± 0.0071	0.7112 ± 0.0095	0.9283 ± 0.0022	9.96

Observations

- **HistGradientBoosting** achieved the highest **Precision**, which was the primary evaluation metric.
- **LightGBM** produced the highest **F1 Score** and **ROC AUC** while performing almost identically to HistGradientBoosting.
- **Random Forest** was the weakest performer despite requiring the longest execution time.
- The boxplots of the five cross-validation folds showed consistent performance across folds, indicating stable model behaviour.

3. Statistical Comparison

Since **HistGradientBoosting** and **LightGBM** were the two strongest models, they were compared using paired statistical tests.

Test	Statistic	p-value
Paired t-test	1.3484	0.2488
Wilcoxon Signed-Rank Test	3.0000	0.3125

Interpretation

Both p-values are greater than **0.05**, so the null hypothesis cannot be rejected.

Although HistGradientBoosting achieved a slightly higher mean Precision than LightGBM, the difference was **not statistically significant**. Therefore, the observed improvement is likely due to normal variation across the cross-validation folds rather than a meaningful performance difference.

4. Engineered Feature Analysis

Feature importance was examined using **Random Forest**, while **Logistic Regression** coefficients were used to understand the direction and strength of individual engineered features.

The most influential engineered features were:

- **education_hours_interaction**
- **log_capital_gain**
- **higher_education**
- **has_capital_gain**
- **log_capital_loss**

These findings are consistent with expectations. Capital gain is a well-known indicator of higher income in the Adult Income dataset, while combining education level with working hours provides additional predictive information that neither feature captures independently.

5. Feature Selection

A feature selection experiment was performed using **SelectKBest (Mutual Information, k = 30)**.

Metric	Full Features	SelectKBest
Precision	0.7461 ± 0.0062	0.7492 ± 0.0070
F1 Score	0.6765 ± 0.0101	0.6589 ± 0.0063
ROC AUC	0.9131 ± 0.0019	0.9076 ± 0.0017
Training Time	11.15 s	56.45 s

Although feature selection produced a very small improvement in Precision, it reduced both F1 Score and ROC AUC while increasing training time by approximately **5×**.

Recommendation for Hyperparameter Tuning

Based on today's experiments:

- **Primary model:** HistGradientBoosting
- **Alternative model:** LightGBM
- **Features:** Retain the complete engineered feature set instead of applying SelectKBest.

The complete feature set provides the best balance between predictive performance and computational efficiency. Although SelectKBest slightly improved Precision, the gain was minimal and came at the cost of lower F1 Score, lower ROC AUC, and significantly longer training time. Consequently, the full engineered feature set is the preferred choice for tomorrow's hyperparameter tuning.